Mathematics Formula Book

Important formulas and quick revision guide

Algebra & Complex Numbers

• Quadratic roots: $x = (-b \pm \sqrt{(b^2-4ac)})/(2a)$.

• Discriminant: $D = b^2-4ac$.

• Complex modulus: $|a+ib| = \sqrt{(a^2+b^2)}$.

• $(a+ib)(a-ib)=a^2+b^2$.

Sequences & Series

• AP nth term: $a_n = a+(n-1)d$.

• AP sum: $S_n = n/2\ [2a+(n-1)d]$.

• GP nth term: $a_n = ar^{(n-1)}$.

• Finite GP sum: $S_n = a(r^n-1)/(r-1)$, for $r\neq1$.

Permutation & Combination

• $n! = n(n-1)(n-2)\ldots1$.

• $nPr = n!/(n-r)!$.

• $nCr = n!/[r!(n-r)!]$.

• $nCr = nC(n-r)$.

Binomial Theorem

• $(a+b)^n = \Sigma\ nCr\ a^{(n-r)}b^r$.

• General term: $T_{(r+1)} = nCr\ a^{(n-r)}b^r$.

Matrices & Determinants

• For [[a,b],[c,d]], determinant = $ad-bc$.

• 2×2 inverse = $1/(ad-bc)$ [[d,–b],[–c,a]], when $ad-bc\neq0$.

• Matrix multiplication requires compatible dimensions.

Coordinate Geometry

• Slope between two points: $(y_2-y_1)/(x_2-x_1)$.

• Distance between points: $\sqrt{[(x_2-x_1)^2+(y_2-y_1)^2]}$.

• Circle: $(x-h)^2+(y-k)^2=r^2$.

• Parabola standard forms include $y^2=4ax$ and $x^2=4ay$.

Calculus

- Derivative: $d(x^n)/dx = n x^{(n-1)}$.
- $d(\sin x)/dx = \cos x$; $d(\cos x)/dx = -\sin x$.
- $\int x^n dx = x^{(n+1)}/(n+1)+C$, $n \neq -1$.
- $\int 1/x \, dx = \ln|x|+C$.

Vectors & 3D

- $|a| = \sqrt{(a \cdot a)}$.
- $a \cdot b = |a||b|\cos\theta$.
- $|a \times b| = |a||b|\sin\theta$.
- Perpendicular vectors have dot product zero.

Probability & Statistics

- $P(A \cup B)=P(A)+P(B)-P(A \cap B)$.
- $P(A|B)=P(A \cap B)/P(B)$, when $P(B) \neq 0$.
- Mean = sum of observations / number of observations.
- Variance is the average squared deviation from the mean.